

CSCI 6917/6918:
Guided Research Methods
3-5 Page Project Proposal
Due Lecture 4 COB

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Project Title: Measuring Gender, Age, and Nationality Bias in Transformer-Based Language Models: A Comparative Study of BERT and RoBERTa

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1. What are you going to do?

The goal of this study is to quantify and contrast demographic biases in BERT and RoBERTa, two popular transformer-based language models. In particular, the project will use the CrowS-Pairs benchmark dataset to examine biases related to gender, age, and nationality. The CrowS-Pairs Bias Score will be used to assess the models' preference for stereotypical vs less stereotypical utterances. Finding variations in bias levels between demographic groups and model designs is the aim. Growing worries about ethics and justice in AI systems are the driving force behind this research. Applications that affect people's access to information, job prospects, healthcare help, and digital communication are increasingly utilizing transformer-based language models. Because these models learn from large-scale human-generated text, they may reproduce harmful stereotypes or unequal treatment toward certain demographic groups.

The project will focus on three specific demographic categories: gender, age, and nationality. These categories are selected because they are socially important, commonly discussed in fairness research, and well represented in the CrowS-Pairs benchmark dataset. The study will evaluate whether BERT and RoBERTa systematically prefer stereotypical statements over less stereotypical alternatives within these categories.

2. How is it done today? Current Limitations?

Benchmark datasets like CrowS-Pairs, StereoSet, and WinoBias are frequently used to assess bias in language models. A model's preference for stereotypical or anti-stereotypical utterances is usually measured by researchers. Current research frequently focuses on a small number of bias categories or evaluates a single model, despite the fact that these benchmarks offer consistent assessment methodologies. Additionally, bias measures might differ based on the assessment technique and dataset, which makes cross-study comparisons challenging. Additionally, nothing is known about how various transformer models compare in terms of certain demographic biases like age and nationality. Many language models include quantifiable social biases, according to existing research, however because various datasets, measurements, and experimental setups are used in different studies, it is sometimes challenging to compare the results. While some studies exclusively look at gender prejudice, others look at more general categories like ethnicity or

religion. Furthermore, a single model is used for numerous assessments, which makes it challenging to comprehend how several designs compare under the same circumstances. Another limitation is that benchmark datasets do not fully capture the complexity of real-world language use. While datasets such as CrowS-Pairs provide controlled sentence pairs for evaluation, they may not represent all possible contexts in which bias can appear. Nevertheless, they remain valuable tools for conducting reproducible and standardized comparisons between models.

3. What is your new idea?

Three demographic bias categories—gender, age, and nationality—will be the focus of the proposed work's comparison between BERT and RoBERTa. The initiative focuses on a manageable selection of categories within a single benchmark dataset instead than trying to address all types of bias. This method offers insights into how model training techniques may impact bias while enabling a controlled and repeatable comparison of two significant transformer models. This project's concentrated and comparative design is what makes it unique. The study compares two powerful transformer models under identical settings and focuses on three well-defined demographic characteristics rather than trying to assess every potential type of bias. The contrasts between BERT and RoBERTa may be interpreted more directly thanks to this regulated arrangement. Reproducibility is another focus of the research. Both models will utilize the same dataset, evaluation metric, preparation procedures, and experimental setup. The goal of this consistency is to lessen methodological variability and make it easier to understand how training methods and model design may affect demographic bias.

4. Who will benefit from your work? Why?

AI academics, machine learning professionals, and companies that use language models in practical applications stand to gain the most. Because biased model outputs can have a detrimental impact on fairness, inclusion, and user trust, it is crucial to understand demographic biases. The results of this study might aid developers in comprehending the shortcomings of popular language models and encourage smarter choices when choosing models for delicate applications. In addition, the results may benefit educators and students studying responsible AI by providing a practical example of how demographic bias can be measured in modern language models. The findings may also encourage more careful evaluation of AI systems before deployment in sensitive environments such as education, recruitment, and customer service.

5. What risks do you anticipate?

One risk is that the CrowS-Pairs dataset may not capture all real-world manifestations of demographic bias. Another risk is that observed differences between models may be relatively small, making interpretation challenging. There is also a possibility that evaluation results may be influenced by implementation details or model versions. To reduce these risks, the project will use standardized evaluation procedures, documented software versions, and reproducible experimental settings. There is also a risk that bias scores alone may not fully explain the practical impact of biased behavior in real-world applications. A model may achieve a moderate bias score while still producing harmful outputs in specific contexts. Therefore, the project will complement

quantitative evaluation with qualitative analysis of representative examples to better interpret the results.

6. Out of pocket costs? Complete within 11 weeks?

The project is expected to have little to no out-of-pocket cost. All required resources, including the CrowS-Pairs dataset, BERT, and RoBERTa models, are publicly available and open source. Experiments can be conducted using freely available computational resources such as Google Colab or university computing facilities. The scope of the project is intentionally limited to two models and one benchmark dataset, making completion within the 11-week course period realistic and achievable. The proposed timeline is feasible because the project scope is intentionally limited. Only two models and one benchmark dataset will be evaluated, which keeps the computational requirements manageable. The use of pre-trained models also eliminates the need for expensive model training from scratch, allowing the project to focus on evaluation and analysis within the available timeframe.

7. Interim Project Report

By the midpoint of the project, the following milestones will be completed:

- Literature review on bias in language models.
- Collection and preparation of the CrowS-Pairs dataset.
- Setup of the experimental environment.
- Implementation of the evaluation pipeline.
- Preliminary evaluation of BERT on selected bias categories.
- Initial analysis of gender, age, and nationality bias results.

The interim report will summarize related work, methodology, implementation progress, preliminary findings, and any challenges encountered. The interim report will also include a preliminary comparison of bias trends across the selected categories and a discussion of any unexpected observations. If necessary, adjustments to the evaluation pipeline or analysis methodology will be documented at this stage.

8. Final Demonstration (August)

The final demonstration will present a complete comparative evaluation of BERT and RoBERTa using the CrowS-Pairs dataset. Results will include quantitative bias scores for gender, age, and nationality categories, along with visualizations and comparative analysis. The demonstration will highlight key findings regarding differences between the models and discuss implications for fairness in transformer-based language models. Deliverables will include the source code, experimental documentation, reproducible evaluation procedures, and a final written report summarizing the research outcomes. The final demonstration will emphasize both technical implementation and research findings. It will include an overview of the evaluation process, visual summaries of bias scores, and examples illustrating how the models respond differently to stereotypical and non-stereotypical statements. The demonstration will conclude with a discussion

of limitations, implications for responsible AI, and possible directions for future work, including bias mitigation techniques.